Neutron Transport and Tritium Breeding Modelling in Nuclear Fusion Reactor Breeder Blankets

MPhys Project Report

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Abstract

The realisation of commercial nuclear fusion via the deuterium-tritium (D-T) fuel cycle is contingent upon achieving tritium self-sufficiency. This project utilises a coupled computational workflow, integrating Paramak for parametric CAD generation and OpenMC for stochastic neutron transport, to optimise the neutronic performance of tokamak breeder blankets. A systematic parameter sweep was conducted to evaluate the impact of material composition, isotopic enrichment, and reactor geometry on the Tritium Breeding Ratio (TBR).

Results indicate that the liquid eutectic $Li_{17}Pb_{83}$ offers superior performance, achieving a maximum TBR of 1.61, significantly outperforming molten salts. Solid ceramic concepts provide more consistent strong TBR performance of 1.48 ± 0.03. Contrary to high-enrichment assumptions, this study identifies an enrichment saturation point at approximately 40% ^{6}Li, beyond which performance diminishes due to the loss of ^{7}Li neutron multiplication. Zirconium alloy first walls were found to be superior for TBR. Geometric optimisation reveals that TBR saturates at a radial blanket depth of 500–600 mm. Under a homogenous approximation of the breeder blanket structure, water coolant was found to enhance breeding via moderation, gaseous coolants (He, N_2) act as voids that degrade the neutron economy.

While these findings suggest several avenues to optimise TBR, informing design constraints for next-generation devices, focusing on TBR optimisation can lead to compromises in other key parameters required for realising fusion like heat extraction. Further work in relaxing the model approximations will provide greater fidelity and applicability for these results.

CONTENTS

1. INTRODUCTION

The global demand for energy is rising inexorably, coinciding with an urgent necessity to decarbonise power generation. In this context, nuclear fusion offers the prospect of a near-limitless, low-carbon energy source. By replicating the stellar nucleosynthesis processes that power the Sun, fusion energy aims to generate electricity without long-lived high-activity radioactive waste or the risk of runaway chain reactions [1].

Among the various reaction pathways, the deuterium-tritium (D-T) fuel cycle is the most viable for near-term applications due to its high reaction cross-section at achievable plasma temperatures [2]. However, while deuterium is abundant in Earth's oceans, the practical realisation of this cycle faces a fundamental logistical challenge: the supply of tritium. Tritium is a radioactive isotope ($\tau_{1/2} \approx 12.3$ years) with no significant natural abundance [3]. Current global inventories are derived as by-products of heavy water moderated fission reactors and are insufficient to support a future fleet of commercial fusion power plants (see Section 3.1).

Consequently, a D-T fusion reactor cannot rely on an external fuel supply; it must breed its own fuel. This leads to the concept of the *Breeder Blanket*—a component designed to intercept the 14.1 MeV neutrons produced in the fusion reaction and transmute lithium into tritium. While the blanket serves integral roles in heat extraction and shielding, its defining function—and the primary figure of merit for this study—is the **Tritium Breeding Ratio (TBR)**: the ratio of tritium produced to tritium consumed.

Because the D-T reaction produces only one neutron per consumption event, this "one-to-one" neutron economy would require breeding efficiency (no losses of neutrons / tritium). Real world breeding inefficiencies (see Section 3.2) necessitate achieving a TBR greater than 1.0. As the source of fuel for fusion, tritium breeding is not merely a design objective but a threshold requirement for the viability and scalability of fusion energy [4].

This strict margin necessitates high-fidelity modelling. Experimental validation of full-scale reactor neutronics is generally unfeasible, placing the burden of design on computational simulation. This project applies advanced Monte Carlo transport methods to perform a systematic optimisation of the breeder blanket. By utilising a parametric computational workflow, this study aims to identify the specific material compositions and geometric configurations required to maximise the TBR, ensuring fuel self-sufficiency while maintaining adequate performance for other aspects of fusion energy generation. The outcomes of this work are intended to inform the design of next-generation concepts relevant to future magnetic confinement devices, such as the EU-DEMO (DEMOnstration power plant) [5] and the UK STEP (Spherical Tokamak for Energy Production) programme [6].

2. THEORY

2.1. Fundamentals of Nuclear Fusion

Nuclear fusion is the process by which two light atomic nuclei combine to form a heavier nucleus, releasing energy due to the increase in total nuclear binding energy [1]. While various light nuclei interactions can release energy via this mass defect, the deuterium-tritium (D-T) reaction has emerged as the most viable for near-term applications, partially due to historical precedent (see Section 3.1) but mainly due to its physical accessibility.

This reaction is favoured because it possesses a significantly higher fusion cross-section (σ_f) at achievable plasma temperatures (10–20 keV) compared to alternatives such as D-D or D-^{3}He (see Figure 1) [2]. The reaction proceeds as:

$$^2\mathrm{H} + {}^3\mathrm{H} \longrightarrow \mathrm{n} + \alpha + 17.6\,\mathrm{MeV} \tag{1}$$

The release of 17.6 MeV is partitioned inversely to the mass of the products to conserve momentum. Consequently, the electrically neutral neutron carries the majority of the energy ($E_n = 14.1$ MeV), while the charged alpha particle (α) retains 3.5 MeV [2].

In Magnetic Confinement Fusion (MCF) devices, such as the tokamak [7], the charged α-particle is confined by the magnetic field, depositing its energy to sustain the plasma temperature (self-heating). The 14.1 MeV neutron, being electrically neutral, escapes the magnetic confinement. The reactor breeder blanket is the subsystem designed to intercept this neutron flux to extract thermal energy, shield the external components, and breed fuel [8].

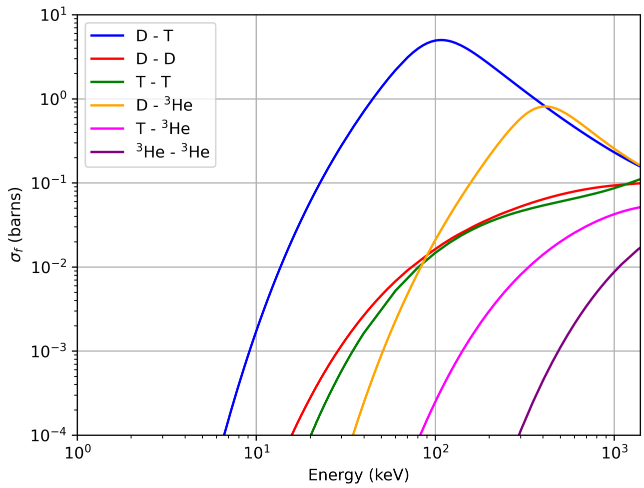

Figure 1: A comparison of fusion reaction reactivities. The total cross-section is plotted as a function of ion temperature (keV) for the D-T, D-D, and D-^{3}He fuel cycles. The D-T reaction peaks at a lower temperature and higher cross-section than alternatives [2].

2.2. Tritium Breeding and Neutron Multiplication

The necessity of tritium production was established in Section 1. To achieve this, the blanket must breed tritium via neutron capture interactions with lithium. Natural lithium consists of two isotopes: ^{6}Li and ^{7}Li (1.9-7.8% ^{6}Li abundance) [3].

2.2.1. Breeding Reactions

Both isotopes undergo tritium-producing reactions, but their neutronic characteristics differ fundamentally:

$$^6\mathrm{Li} + \mathrm{n} \longrightarrow \alpha + \mathrm{t} + 4.8\,\mathrm{MeV} \tag{2}$$

$$^7\mathrm{Li} + \mathrm{n} \longrightarrow \alpha + \mathrm{t} + \mathrm{n}' - 2.5\,\mathrm{MeV} \tag{3}$$

The ^{6}Li reaction (Eq. 2) is exothermic and possesses a $1/v$ cross-section (σ) dependence, making it especially effective at thermal neutron energies ($E < 0.5$ eV) (Figure 2) [9]. Conversely, the ^{7}Li reaction (Eq. 3) is endothermic with a threshold energy of approximately 2.5 MeV [8], and has a cross section orders of magnitude lower. But in equation 3 ^{7}Li releases a lower energy neutron, so does not deplete neutron flux like ^{6}Li.

2.2.2. Neutron Multiplication and Economy

The reactor's fuel self-sufficiency is quantified by the **Tritium Breeding Ratio**, defined as the ratio of tritium bred to tritium consumed.

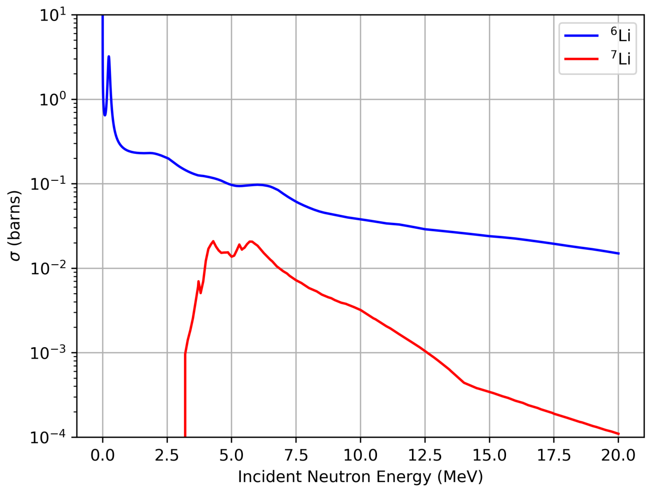

Figure 2: Neutron cross-sections for the primary tritium breeding reactions, ^{6}Li(n,α)t and ^{7}Li(n, n′α)t, as a function of incident neutron energy. Note the dominance of ^{6}Li at thermal energies. Data aquired from ENDF [3]

However, the D-T reaction produces exactly one neutron, and the primary ^{6}Li breeding reaction consumes one neutron. This 1-to-1 economy leaves no margin for breeding inefficiency (see Section 3.2). To overcome this deficit, the blanket must utilise neutron multipliers—materials with significant "spallation" $(\mathrm{n}, 2\,\mathrm{n})$ cross-sections at 14.1 MeV. The primary candidates are beryllium and lead (see Section 3.3):

$$^{9}\mathrm{Be} + \mathrm{n} \longrightarrow {}^{8}\mathrm{Be} + 2\,\mathrm{n}' - 1.57\,\mathrm{MeV} \tag{4}$$

$$^{208}\mathrm{Pb} + \mathrm{n} \longrightarrow {}^{207}\mathrm{Pb} + 2\,\mathrm{n}' - 7.37\,\mathrm{MeV} \tag{5}$$

These are threshold reactions (see Figure 3) that increase the macroscopic neutron population, allowing a single incident fusion neutron to potentially breed more than one triton [9]. Beryllium has a much lower cross section, but is able to spallate at lower neutron energies (∼2.5MeV). Though lead requires higher neutron energies (∼8MeV), it has a significantly higher cross section.

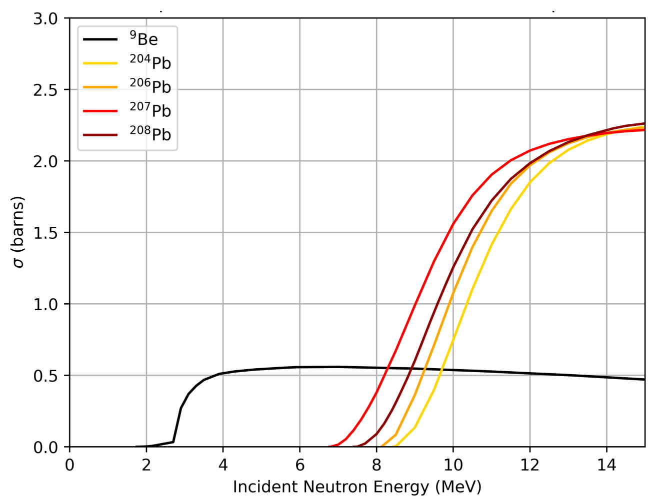

Figure 3: Neutron cross-sections for the $(\mathrm{n}, 2\,\mathrm{n})$ multiplication reactions in beryllium and lead. These threshold reactions effectively double the neutron population available for breeding. Data aquired from ENDF [3]

2.3. Neutron Transport and Interactions

To determine the TBR, one must model the full phase-space behaviour of neutrons within the complex reactor geometry. This behaviour is governed by the Boltzmann Transport Equation (BTE).

2.3.1. The Boltzmann Transport Equation

The BTE is a linear integro-differential equation describing the conservation of neutrons in phase space $(\vec{r}, \hat{\Omega}, E, t)$. It stipulates that the rate of change of neutron density is equal to the net production minus net loss and net leakage. For a steady-state system with a fixed source, the BTE is given by [10]:

$$\underbrace{\hat{\Omega} \cdot \nabla\psi}_{\text{Streaming}} + \underbrace{\Sigma_t\psi}_{\text{Interaction Loss}} = \underbrace{\int_0^{\infty} \int_{4\pi} \Sigma_s(E' \to E, \hat{\Omega}' \to \hat{\Omega})\psi' d\hat{\Omega}' dE'}_{\text{Backscattering Source}} + \underbrace{S(\vec{r}, \hat{\Omega}, E)}_{\text{Fixed Source}} \tag{6}$$

Where $\psi(\vec{r}, \hat{\Omega}, E)$ is the angular neutron flux, Σ_t is the total macroscopic cross-section, and Σ_s represents the probability of a neutron scattering from energy E' and direction $\hat{\Omega}'$ into E and $\hat{\Omega}$.

Analytic solutions to Eq. 6 are generally impossible for heterogeneous 3D reactor geometries. Consequently, this project employs Monte Carlo (MC) methods. MC does not solve the equation for average flux directly; instead, it simulates the stochastic histories of individual particles, sampling interaction probabilities from nuclear data libraries such as ENDF/B-VIII.0 [3]. The statistical aggregation of these histories converges to the solution of the integral BTE [10] (see Section 4.1).

2.3.2. Interaction Probabilities, Mean Free Path and Moderation

The probability of a specific interaction type x (e.g., capture, fission, scatter) occurring is defined by the macroscopic cross-section Σ_x, which relates to the microscopic cross-section σ_x and atomic density N:

$$\Sigma_x(E) = N\sigma_x(E) \tag{7}$$

The mean free path (λ), representing the average distance a neutron travels through the blanket medium before interacting [1], is:

$$\lambda(E) = \frac{1}{\Sigma_{total}(E)} \tag{8}$$

In blanket design, λ dictates the necessary radial build thickness required to attenuate the 14.1 MeV flux and protect against neutron leakage [11].

Fast fusion neutrons can be moderated, often all the way to thermal energies, using appropriate low mass number materials (such as water). This is detailed further in Appendix A. This maximises the cross section of the breeding reaction 2, and can reduce neutron leakage by attenuating neutron flux to thermal energies.

3. LITERATURE REVIEW

3.1. The Tritium Fuel Imperative

Given the physical accessibility of DT reactions (see Section 2.1), they have been the front runner for fusion since the post war boom of nuclear research; attempting to replicate the success of fission [12]. As referenced in Section 1, tritium supply is critical constraint on the realisation of DT fusion. Current global tritium stockpiles are derived almost exclusively as a by-product of heavy-water moderated fission reactors (CANDU, CANada DeUterium Reactors)

[13]. Analyses of this supply chain indicate that stockpiles are insufficient to support the operation of a future fusion fleet [13, 14]. The testing and startup of a large scale fusion device like ITER alone is currently predicted to require ∼12.3kg, and the total production of tritium per is (in optimistic scenarios) capped at 8kg per year. This ignores the consumption estimates of operational fusion reactors which range from 10-100kg of tritium per year [6]. Consequently, the external supply chain is considered a bottleneck that threatens the viability of the entire D-T fusion concept. To resolve this, commercial reactors are required to breed their own fuel in-situ via the **Tritium Breeder Blanket**.

3.2. Breeder Blanket Evolution and Functions

The concept of a "breeder" blanket predates magnetic confinement fusion; it was originally conceived to utilise high-energy neutrons to produce fissile materials (^{233}U, ^{239}Pu) for fission warheads and reactors [15, 16, 17]. Modern fusion research has repurposed this architecture to focus on the transmutation of lithium into tritium. A "blanket" refers to a structure that lines the walls of a fusion reactor; "breeder" indicates that tritium is produced upon interacting with neutrons. Beyond breeding, the blanket must simultaneously perform three conflicting functions: shielding superconducting magnets and external components from neutron damage / radiation, extracting high-grade heat for power conversion, and maintaining a TBR > 1.0 [8]. Fusion projects across the world are already outlining blanket designs as a key strategic focus, some examples include ITER's DEMO [18], UK's STEP [6] and the Chinese Fusion Engineering Test Reactor Facility (CFETR) [19].

TBR must be > 1.0 give margin for several different sources of inefficiency. Neutrons can be lost to structural materials (e.g. first walls) via parasitic absorption and neutron can also leak out of the reactor vessel. Produced tritium can be lost from processing inefficiencies and tritium's short half life leads to decay losses. Accounting for these breeding inefficiencies literature typically suggests a TBR significantly > 1.0 (typically ≈ 1.1) [8, 4, 20, 18].

3.3. Neutron Multiplication Strategies

Since natural lithium cannot sustain a TBR > 1 in a realistic reactor geometry solely via the endothermic $^{7}\mathrm{Li}(\mathrm{n}, \mathrm{n}'\alpha)\mathrm{t}$ reaction. Design studies have converged on two primary multiplier candidates due to their high spallation cross section: beryllium and lead [9].

Beryllium (^{9}Be) can be considered superior due to its low threshold energy for the $(\mathrm{n}, 2\,\mathrm{n})$ reaction. It is the multiplier of choice for solid ceramic breeder concepts (e.g., HCPB, WCPB) and molten fluoride salts (FLiBe) [21, 11, 22]. However, its application is controversial due to resource scarcity and high toxicity [23]. This highlights the handling of toxic beryllium dust poses severe safety and manufacturing challenges, prompting research into minimizing its volume fraction in blanket designs.

Conversely, lead (Pb) is favored for higher $(\mathrm{n}, 2\,\mathrm{n})$ cross section, where its threshold energy is typically met for DT fusion ($E_n \sim 14.1$). It also has chemical compatibility with liquid breeders specifically in the lithium-lead eutectic ($\mathrm{Li}_{17}\mathrm{Pb}_{83}$). This alloy allows the medium to function simultaneously as breeder, multiplier, and coolant [24]. While lead mitigates the toxicity concerns of beryllium, it introduces neutronic penalties; it acts as a high-Z shield, parasitically absorbing the thermal neutrons required for efficient ^{6}Li breeding. Furthermore, the activation of lead can generate long-lived radioisotopes, complicating decommissioning [25]. The selection of a multiplier therefore represents a trade-off between the superior neutronic efficiency of beryllium and the integrated engineering advantages of lead [9].

3.4. Divergence in Blanket Architectures

Blanket designs have historically diverged into two competing pathways—liquid and solid—driven primarily by the phase of the breeding medium and secondarily by the choice of coolant (water vs. helium).

3.4.1. Liquid Breeder Concepts

Liquid breeders offer the distinct advantage of continuous tritium extraction and simplified fuel replenishment [11]. The most common liquid breeder is the eutectic alloy lithium-lead (Li−Pb). These designs generally require a dedicated coolant loop to manage thermal loads:

- **WCLL and HCLL:** The water-cooled lithium-lead (WCLL) [18] and helium-cooled lithium-lead (HCLL) [24, 26] concepts are primary candidates for ITER and also investigated for the CFETR. In these designs, the liquid metal serves primarily as a breeder and neutron multiplier, while heat extraction is handled by the separate coolant system. Some nitrogen coolant variants also exist [27].
- **DCLL:** A more advanced variant, the dual-cooled lithium-lead (DCLL) blanket, is a mature concept for the European DEMO [28]. Here, the flowing liquid metal contributes significantly to cooling alongside a helium loop, increasing overall thermal efficiency [29].

However, the flow of liquid metal through the tokamak's strong magnetic field induces significant magnetohydrodynamic (MHD) drag, which can inhibit circulation [30, 31].

3.4.1.1. Molten Salts:

To mitigate these MHD issues, designs such as the ARC reactor utilise molten salts like FLiBe ($2\,LiF \cdot BeF_2$) which possess low electrical conductivity [22, 32]. While this solves the magnetic drag issue, these salts introduce new challenges regarding chemical corrosion and complex flow chemistry control [25, 33].

3.4.2. Solid Breeder Concepts

Solid breeders utilise lithium ceramics (e.g., Li_2TiO_3, Li_4SiO_4) in packed pebble beds combined with distinct coolant architectures. This approach eliminates MHD effects entirely, allows for easier tritium extraction and ensures chemical stability [34, 35].

The choice of coolant dictates the blanket structure and affects neutron transport efficiency:

- **HCPB:** Helium-cooled pebble bed, a primary focus of the DEMO system [21, 36].
- **WCPB:** Water-cooled pebble bed, the CFETR's prioritised approach [37, 19].

While mechanically stable, the primary drawback of solid breeders lies in the complexity of tritium extraction; tritium produced within the ceramic lattice must diffuse out and be purged by a sweep gas, a process sensitive to radiation-induced trapping [38].

3.5. Neutronics Modelling and the Research Gap

Optimising these complex designs requires high-fidelity modelling, as experimental validation of full-scale blanket neutronics is currently impossible [39]. Consequently, the field relies on Monte Carlo (MC) transport codes (e.g., OpenMC, MCNP) to resolve the stochastic behaviour of neutrons in 3D geometries [40, 41].

However, a critical review of the literature reveals a methodological gap in how these tools are applied. Previous studies generally fall into one of two categories:

1. **Broad Material Surveys:** Reviews such as [9] provide excellent "first principles" comparisons of many materials but rely on simplified 1D or homogenised models. These studies sacrifice geometric accuracy for breadth, failing to capture spatial transport effects like streaming or leakage.

2. **Specific Point-Designs:** Studies focused on specific reactors (e.g., ARC [42] or STEP [27]) utilise high-fidelity 3D CAD models but fix the geometry to a single design point. These studies answer "What is the performance of *this* reactor?" rather than "How does geometry influence performance?"

This dichotomy exists because manually generating unique 3D CAD models for a parametric sweep has historically been computationally intractable. Recent developments in parametric CAD generation have begun to bridge this gap. Goel et al. [43] demonstrated the viability of automated geometry generation for neutronic sweeps. However, their study—representing an initial step in this domain—did not fully exploit the coupling between material composition and geometric variance.

This project aims to close this gap by applying an automated, parametric workflow to a realistic 3D tokamak model. Unlike previous studies which isolated material choice from geometric design, this work will conduct a systematic parameter sweep of variables—including varying ^{6}Li enrichment, blanket thickness, and material ratios. This approach allows for the identification of optimal design points that lie at the intersection of material science and reactor geometry, regions of the design space that effectively remain unmapped in current literature.

4. METHODOLOGY

4.1. Modelling Neutron Transport: Monte Carlo

The neutronic performance of the breeder blanket is determined by the transport behaviour of the entire neutron population within the reactor geometry under the BTE (equation 6). As explained in Section 2.3.1, the **Monte Carlo (MC) method** is used to solve for neutron transport behaviour. The MC method is a stochastic technique that simulates the transport of neutrons by tracking the individual life histories of discrete particles [44]. Rather than solving for global flux directly, the simulation proceeds through a random walk process:

1. **Source Sampling:** A neutron is initialised at the plasma source with sampled parameters (e.g., 14.1 MeV energy and isotropic direction).

2. **Ray Tracing:** The distance to the next interaction is sampled from a probability distribution derived from the total macroscopic cross-section of the local material [3].

3. **Interaction Physics:** At the collision site, the specific reaction type (elastic scatter, absorption, $(\mathrm{n}, 2\,\mathrm{n})$ multiplication, etc.) is selected probabilistically based on the nuclide-specific cross-section data.

4. **History Termination:** The particle's energy and direction are updated after scattering. The process repeats until the neutron is absorbed or leaks from the system geometry. Multiplication events generate secondary neutrons which are subsequently tracked.

4.1.1. Statistical Uncertainty and Error Analysis

Unlike deterministic solvers, Monte Carlo results represent statistical estimates rather than exact solutions. The physical quantities of interest, such as the TBR, are tallied by averaging contributions across many independent particle histories. The uncertainty in these results is quantified using the standard deviation of the mean across simulation batches.

The relative error (R) in an MC calculation is governed by the Central Limit Theorem and scales inversely with the square root of the number of simulated particles [10, 40] (N):

$$R \propto \frac{1}{\sqrt{N}} \tag{9}$$

To ensure statistical robustness, the simulations in this work utilised high particle counts (typically $N \sim 10^5 - 10^7$) to drive convergence. This resulted in relative errors significantly below 1%.

4.2. Computational Workflow and Toolchain

As established in Section 3, modern neutronics research relies on a robust toolchain linking geometric modeling with particle transport simulation. This project employs a coupled workflow utilising two primary open-source tools: **Paramak** for geometry generation and **OpenMC** for neutron transport.

4.2.1. Geometry Generation: Paramak

To manage the complex toroidal geometry of the tokamak, this project utilises **Paramak**, a Python-based parametric CAD framework [45]. Paramak allows for the automated generation of tokamak fusion reactor models through a high-level API, enabling the rapid modification of geometric features—such as blanket thickness, triangularity, and radial build—without manual CAD reconstruction.

The generated geometry is exported as a DAGMC (Direct Accelerated Geometry Monte Carlo) unstructured mesh. This format is optimised for ray-tracing operations, allowing the Monte Carlo code to transport particles directly on the CAD surface representation without voxelization.

4.2.2. Neutron Transport: OpenMC

All neutronic simulations are performed using **OpenMC**, a state-of-the-art Monte Carlo transport code [40]. OpenMC is widely adopted within the fusion community due to its continuous-energy physics and efficient parallelisation. The code has been extensively validated for complex fusion applications, demonstrating high fidelity in calculating critical parameters such as the TBR and volumetric nuclear heating [41].

To ensure computational efficiency while maintaining physical accuracy, the simulations utilise a 90° sectoral revolution of the tokamak with reflective periodic boundary conditions applied to the radial surfaces (Figure 4).

4.3. Definition of the Base Reactor Model

Before undertaking parametric optimisation, a "Base Model" was established to serve as a reference point. The parameters for this baseline were synthesised from literature to ensure the reactor possesses dimensions and material compositions representative of a realistic DEMO-class machine. A summary of these base parameters is provided below.

4.3.1. Neutron Source Specification

The plasma source is modelled as an isotropic, 14.1 MeV neutron ring source located at the magnetic axis. To approximate the Doppler broadening resulting from the thermal motion of plasma ions, the energy spectrum follows a Muir distribution [46]. The ion temperature is set to 20 keV, consistent with burning plasma conditions described by Wandelt [47]. Note that the plasma is treated solely as a neutron ring source; a physical plasma volume with self and wall interactions are outside the scope of this neutronic study.

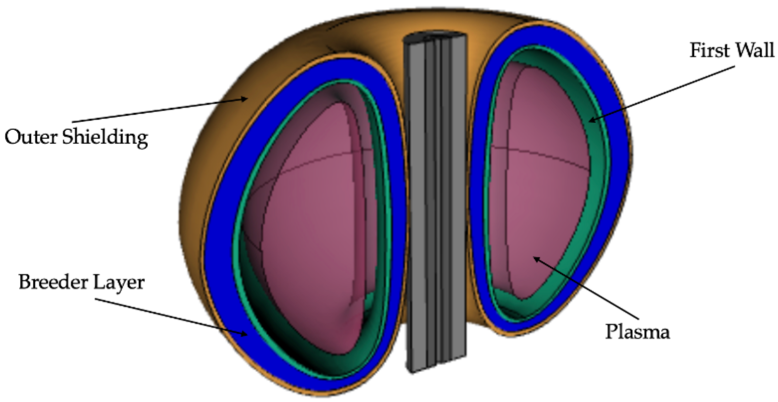

Figure 4: Representative geometry of the simplified Tokamak model generated via Paramak. The simulation domain comprises a 90° revolution with periodic boundary conditions to approximate the full torus while reducing computational cost.

4.3.2. Breeder and Coolant Composition

The reference liquid breeder material is the eutectic lithium-lead alloy ($Li_{17}Pb_{83}$). This specific stoichiometry is selected for its low melting point, requiring less power to melt and mitigating freezing concerns in the loop [48]. There is no hard consensus regarding Li_6 enrichment, literature varies from a low as 10% all the way to 90% [9, 49, 32, 42, 50]. To take an average values of the most common values found in breeding studies, the ^6Li enrichment is initialised at 60%.

Pressurised liquid water (H_2O) is selected as the base coolant due to its prevalence in designs such as the ITER TBM and DEMO concepts [51], with helium and nitrogen coolants considered (Section 3.4). To maintain computational tractability, the breeder and coolant are modelled as a homogenised mixture rather than discrete cooling channels. The coolant occupies 33% of the total blanket volume, as an estimation of piping volume of typical setups [21].

For the comparative analysis of solid ceramic breeders, a specific multiplier configuration was defined to ensure adequate neutron population, informed as a design common in a breadth of older and more recent literature [8, 21, 35]. These simulations utilised a titanium-beryllide multiplier ($Be_{12}Ti$) mixed with the ceramic breeder at a Breeder-to-Multiplier volume ratio of 1:3. To accurately approximate the macroscopic density of a ceramic pebble-bed concept, a packing factor of 62% was applied to the material definition, accounting for the inter-particle void space inherent to packed beds.

4.3.3. Reactor Geometry and First Wall

The first wall (FW) material is zirconium-alloy-4 (Zr-4: 98.2% Zr, 1.5% Sn, 0.2% Fe, 0.1% Cr). Zirconium is selected over standard ferritic-martensitic steels for the base model due to its significantly lower neutron absorption cross-section [52]. The FW thickness is set to 25 mm, an average value derived from structural integrity studies [49, 37].

The breeder blanket depth is set to an average of 500 mm, consistent with modular designs found in literature [37, 19]. The reactor triangularity is fixed at the preset 0.55 in the base model, which is a parameter that controls how triangular the cavity is for plasma distortion which can improve plasma performance in real tokamaks [7].

4.4. Simulation Strategy and Parametric Sweeps

To explore the design space surrounding the Base Model, this project implements a custom, automated simulation pipeline. Leveraging the Python interfaces of both Paramak and OpenMC, an iterative script varies specific design parameters while holding the remaining Base Model attributes constant.

The workflow for each parametric sweep follows this sequence:

1. **Parameter Injection:** A target variable (e.g., Blanket Thickness, ^{6}Li Enrichment, or Coolant Fraction) is modified in the input dictionary.
2. **Geometry Reconstruction:** Paramak regenerates the 3D CAD model and converts it to the DAGMC h5m format.
3. **Material Assignment:** OpenMC applies the material definitions, updating isotopic fractions (e.g., adjusting the lead-lithium ratio) as required by the sweep.
4. **Transport Simulation:** The stochastic transport simulation is executed, tallying neutron interactions in the blanket volume to determine the TBR.

This methodology allows for the isolation of individual variable effects, providing clear trends on how geometric and material changes influence the overall neutronic performance.

5. RESULTS

5.1. Material Composition and Performance

The neutronic performance of candidate liquid and solid breeder materials is summarised in Table 1 and Table 2, respectively.

Table 1: A summary of liquid breeder materials, their properties, and simulated TBR (error on each ≤ 0.005) and simulated average heating per neutron (error on each ≤ 0.05). Materials above the dashed line are the primary materials considered for breeder blankets. Material data was obtained from [9], [42] and [49].

Material	Ratio (molar %)	Li (Atomic %)	ρ (g/cm^3)	TBR	Heating (MeV)
$Li_{17}Pb_{83}$	-	17.00	11.00	1.61	11.5
$LiF-BeF_2$	2:1	28.57	2.04	1.15	12.9
$LiF-BeF_2$*	1:1	20.00	2.06	1.16	13.0
$LiF-PbF_2$	2:3	15.38	3.55	1.23	11.6
LiF	-	50.00	2.64	1.21	13.5
Li	-	100	0.51	1.17	13.6
Li_3N	-	75.00	1.30	1.13	13.3
$LiF-BeF_2-NaF$	1:1:1	14.29	2.15	1.09	12.4
$LiF-NaF-ZrF_4$	55:22:23	20.45	2.72	1.07	11.6
$LiF-LiI_2$	83.5:16.5	50.00	3.68	1.02	11.0
$LiF-LiBr-NaF$	14:79:7	45.69	3.20	0.98	15.6
$LiF-LiBr-NaBr$	20:73:7	46.50	3.16	0.97	15.6
$LiCl-PbCl_2$	1:2	12.50	4.50	0.96	9.9
$LiCl-NaCl$	72:28	36.00	1.52	0.91	11.0
$LiF-NaF-KF$	93:23:84	20.17	2.02	0.90	11.3
$LiCl-BeCl_2$	1:1	20.00	1.52	0.88	10.9
$LiCl-KCl$	7:3	35.00	1.51	0.85	10.7
$LiCl-KCl$*	52:48	26.00	1.51	0.80	10.4

Among the liquid breeders, the eutectic lithium-lead alloy ($Li_{17}Pb_{83}$) demonstrates the highest TBR of 1.61. While high TBR typically correlates with increased nuclear heating due to the exothermic ^{6}Li reaction, $Li_{17}Pb_{83}$ is a notable outlier.

Table 2: A summary of solid breeder materials, their properties, and simulated TBR (error on each ≤ 0.005) and simulated average heating per neutron (error on each ≤ 0.05). Material data was obtained from [9] and [19], where we applied a packing factor of 62% on these quoted densities for our simulations to account for actual occupied space.

Material	Li (Atomic %)	ρ (g/cm^3)	TBR	Heating (MeV)
Li_4SiO_4	44.44	2.40	1.47	14.9
Li_2TiO_3	33.33	3.43	1.46	14.6
Li_8PbO_6	53.33	4.28	1.51	15.0
Li_8CeO_6	53.33	3.25	1.51	15.3
Li_8ZrO_6	53.33	2.98	1.50	15.0
Li_8SnO_6	53.33	3.41	1.49	15.2
$Li_6Zr_2O_7$	40.00	3.56	1.49	14.8
Li_8GeO_6	53.33	2.64	1.49	15.0
Li_5FeO_4	50.00	2.64	1.48	14.9
Li_5AlO_4	50.00	2.25	1.48	15.0
Li_8SiO_6	53.33	2.20	1.48	15.0
Li_6ZnO_4	54.55	2.86	1.48	15.0
Li_4TiO_4	44.44	2.57	1.48	14.9
Li_6MnO_4	54.55	2.50	1.48	14.9
Li_4GeO_4	44.44	3.16	1.48	14.8
Li_2O	66.66	2.10	1.47	14.9
Li_2SiO_3	33.3	2.53	1.45	14.8
Li_2MnO_2	40.00	3.90	1.43	14.4
Li_8CoO_6	53.33	2.47	1.43	14.7
Li_6CoO_4	54.55	2.77	1.41	14.6

Table 2 indicates that solid ceramics breeders display remarkably consistent performance compared to liquids, demonstrated by the low standard deviation on the average TBR ($= 1.48 \pm 0.03$).

5.2. Isotopic and Compositional Optimisation

Figure 5 illustrates the sensitivity of TBR to ^{6}Li enrichment. The dip in TBR at high percentages is attributed to the depletion of ^{7}Li, eliminating the contribution of $^7\mathrm{Li}(\mathrm{n}, \mathrm{n}'\alpha)\mathrm{t}$ reactions which are valuable for preserving the neutron population in the high-energy spectrum.

The simplified study of the lead multiplier fraction in LiPb breeders is shown in Figure 6, where an approximation was taken where the density is calculated as a simple weighted average of Li and Pb. Though there was a peak at 50% lead composition, this TBR was still far lower than the TBR from $Li_{17}Pb_{83}$.

5.3. Impact of Coolant Selection

The influence of coolant fraction and type on breeding efficiency is detailed in Figure 7. For helium and nitrogen, TBR decreases monotonically as the coolant fraction increases, and overlaps to the sweep where we just replaced the coolant with a void. This behaviour indicates that these gases effectively act as voids within the breeder blanket, displacing active breeding material.

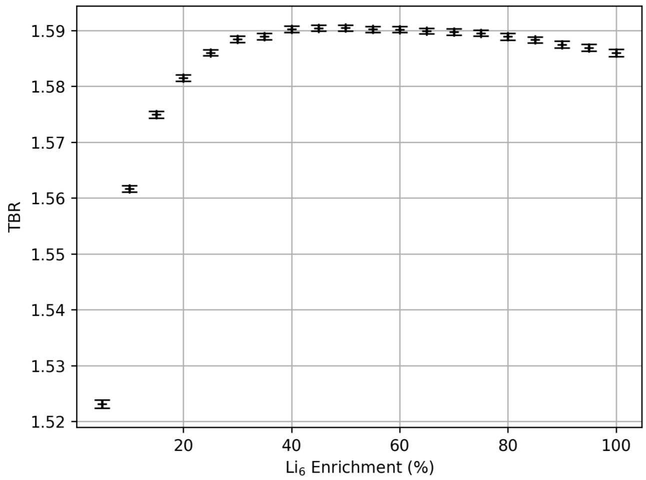

Figure 5: TBR as a function of ^{6}Li enrichment percentage (starting at 5%). The curve is characterised by a rapid initial increase which saturates early; the "knee" of the curve appears at approximately 20%, with the TBR effectively capping at 40%. Beyond this saturation point, the returns are diminishing. Notably, a slight suppression in breeding performance is observed at near-total enrichment levels ($> 90\%$)

Conversely, water (H_2O) exhibits a non-monotonic trend where TBR initially increases before declining. Notably the TBR was the very similar at 40% water coolant as when no coolant was simulated.

5.4. Geometric Parameter Sweeps

Results from the geometric optimisation studies indicate that reactor triangularity has a negligible impact on global neutronics (Figure 8).

The first wall thickness has a significant impact on performance. Figure 9 shows a linear decrease in TBR as first wall thickness increases, corresponding to expected neutron flux attenuation from macroscopic cross section (Section 2.3.2). The comparison reveals that zirconium-4 alloy provides superior breeding performance.

Finally, Figure 10 determines the optimal radial depth of the breeder blanket. TBR increases rapidly with thickness before saturating in the 400–600 mm range. This saturation suggests that the neutron flux is fully attenuated beyond this depth, a conclusion supported by the reaction density heatmap (Figure 10). Increasing the breeder depth beyond 500 mm yields negligible gains in tritium production.

6. DISCUSSION

The results presented in Section 5 quantify the neutronic performance of various breeder blanket concepts. While the simulation data identifies specific maxima for TBR, the selection of a viable blanket design for a fusion power plant requires a holistic evaluation that balances neutronic efficiency against thermal-hydraulic, structural, and economic constraints [9, 5].

6.1. TBR Optimisation Strategies

6.1.1. Isotopic Optimisation

A significant finding is the behaviour of ^{6}Li enrichment (Figure 5). While high enrichment is often assumed to be beneficial, our results indicate a saturation point at approximately 40% enrichment. Beyond this point, the blanket becomes "neutron-limited" rather than "target-limited." Furthermore, enrichment levels approaching 100% actually degrade performance.

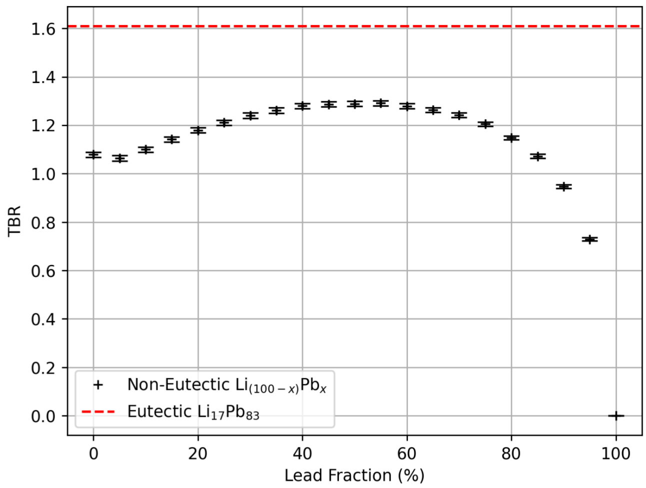

Figure 6: TBR dependence on lead fraction in a simplified non-eutectic / constant packing factor $Li_{(100-x)}Pb_x$ breeder, where the density is calculated as a simple weighted average of Li and Pb. Breeding performance peaks at approximately 50% lead concentration, with a rapid decline observed beyond 75%. The eutectic point is marked for reference, far exceeding any non-eutectic TBR.

This is attributed to the total depletion of ^{7}Li, which removes the contribution of the $^7\mathrm{Li}(\mathrm{n}, \mathrm{n}' \alpha)\mathrm{t}$ reaction [4]. Although this reaction is less frequent, it preserves the neutron population by emitting a secondary neutron, whereas the ^{6}Li reaction is purely absorptive. Economically, this suggests that costly high-enrichment processes are unnecessary; a moderate enrichment of 20–40% represents the optimal trade-off between TBR and fuel cycle costs [43, 32].

6.1.2. Compositional Optimisation

The study confirms that the eutectic lithium-lead alloy ($Li_{17}Pb_{83}$) offered superior breeding performance compared to molten salts, primarily driven by the neutron multiplication capabilities of lead [30]. However, the simulation highlights a critical physical distinction: the superior performance of the eutectic alloy over non-eutectic mixtures (Figure 6) cannot be explained by stoichiometry alone. The eutectic formation allows for a packing efficiency that results in a density approximately 15% higher than the weighted average of its constituents [48]. This increased macroscopic cross-section is the dominant factor in maximising the neutron economy.

Molten salts generally resulted in lower TBRs compared to liquid metals but are investigated due to their reduced Magnetohydrodynamic (MHD) effects. Of the halide salts, fluorine-based compounds (e.g., FLiBe) perform best due to their low neutron absorption cross-sections. Bromine salts exhibit high parasitic absorption, leading to significantly higher heating values (approx. 15.5 MeV). Despite this, bromine's moderating properties ensured it still retained higher TBR than chlorine.

Among the other specific solid breeder compounds (Table 2), cerium-based ceramics achieved the highest TBR due to the low parasitic capture cross-sections of cerium. Conversely, cobalt-based compounds act as outliers, yielding lower TBRs despite the high multiplier fraction; this is attributed to the higher absorption and scattering cross-sections of cobalt, which degrade the neutron economy. The average performance of the solid breeders was strong (TBR = 1.48 ± 0.03) and consistent, higher TBR on average than liquid breeders. A large part of this uniformity arises both from the relatively similar atomic compositions of the molecules (mostly lithium and oxygen) but also the standardised multiplier configuration ($Be_{12}Ti$ at a 1:3 ratio) defined in Section 4.3.2. Nonetheless, noting the easier tritium extraction and better heat performance in pebble beds [35] could place solid breeders as more suitable overall.

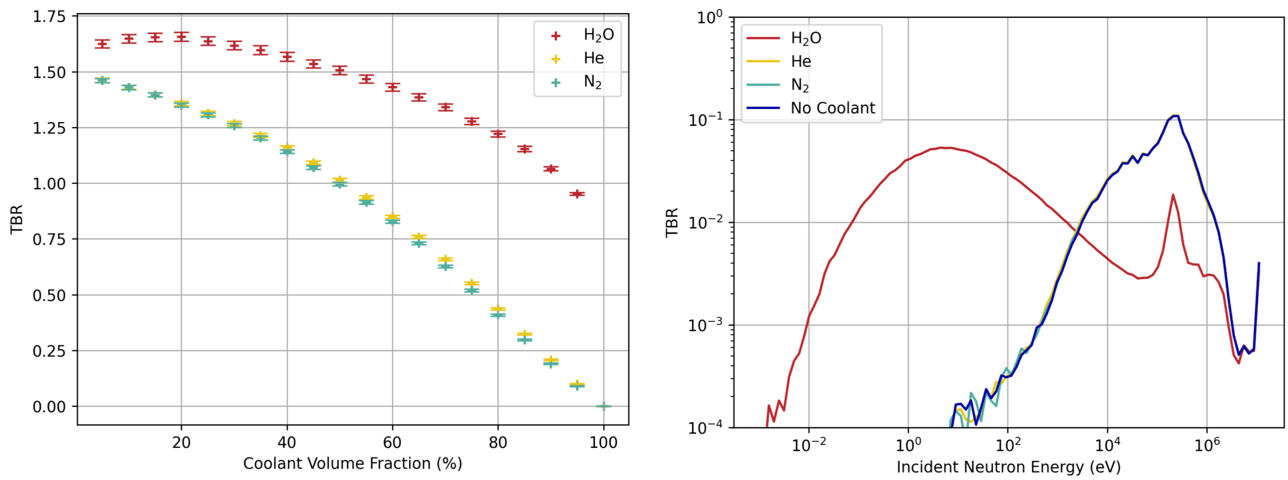

Figure 7: Impact of coolant choice on breeding performance. On the left, TBR vs. fraction of breeder blanket layer replaced with coolant (by volume). Gaseous coolants (He, N_2) act as voids and cause steady reductions in TBR as fractions increase. H_2O, however, has a relatively flat TBR curve up to a fraction of 50%, even increasing TBR at low fractions. The TBR had shallower decreases at high water fractions. On the right, incident neutron energy spectrum for coolant volume fraction = 33%. The H_2O spectrum shows significant moderation compared to the gaseous coolants.

6.1.3. Coolant Optimisation

Water performed the best by a significant margin as a coolant. This is attributed to the moderating properties of water, which thermalises the neutron spectrum. As shown in Figure 7, the water coolant shifts the incident neutron population toward lower energies (Figure 2). Since the tritium production cross-section of ^{6}Li is significantly higher at thermal energies, this moderation compensates for the displacement of breeder volume at lower coolant fractions. This moderation will also reduce the neutron leakage, and so improve heating along with TBR. This then introduces the importance of considering neutron mean free path, flux attenuation and neutron leakage within the blanket, leading us to geometric studies.

6.1.4. Spatial and Geometric Optimisation

The saturation of TBR at a radial depth of roughly 500 mm (Figure 10) can be validated physically by considering the neutron mean free path (equation 8). Dominated by lead, the macroscopic total cross-section (equation 7) of the breeder mixture yields a mean free path of $\lambda \approx 5.5$ cm for 14 MeV neutrons. A 500 mm blanket therefore represents approximately 9 mean free paths, sufficient to attenuate over 99.9% of the incident flux [10]. Increasing the blanket thickness beyond this point provides no neutronic benefit and would act only as dead weight, complicating remote maintenance and increasing capital costs [18].

6.2. Consequences to Other Critical Criteria

While this study maximised TBR, the "optimal" neutronic configurations often conflict with other critical reactor criteria.

6.2.1. Structural Integrity vs. Neutronics

The use of zirconium-4 (Zr-4) as the first wall material yielded the highest TBR due to its low parasitic absorption cross-section (Figure 9) [52]. However, Zr-4 is generally unsuitable for the extreme neutron fluence and thermal stress of a DEMO-class reactor first wall [52]. This also highlights that, although Figure 9 suggests setting the first wall as thin as possible, as heat constraints must be considered to limit how thin it can be [53]. Standard Reduced Activation

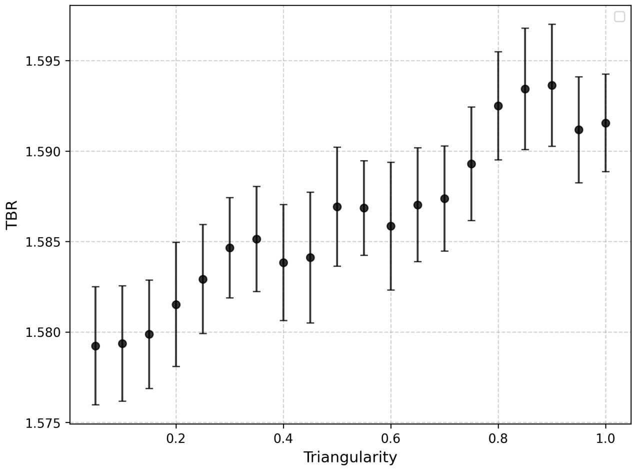

Figure 8: Sensitivity of TBR to reactor triangularity (δ). Noting the y-scale, the gradient of 0.015 ± 0.001 indicates that triangularity has a negligible impact on global neutronics.

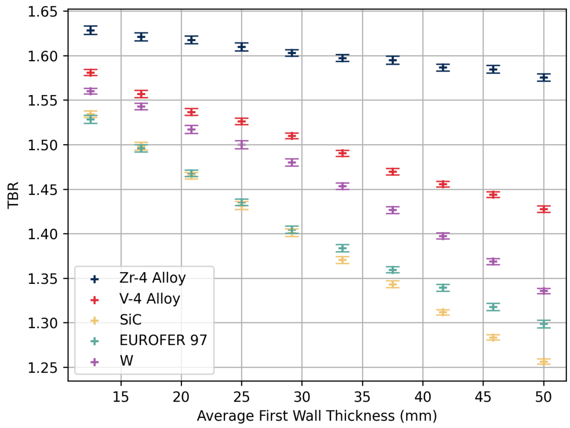

Figure 9: TBR degradation as a function of first wall thickness for various candidate materials, including EUROFER97 [53], Zr-4 as defined in Section 4.3.2 and the rest defined in [52]. Zirconium-based alloys demonstrate the highest TBR profile.

Ferritic-Martensitic (RAFM) steels, such as EUROFER 97, are the engineering standard despite their higher neutron penalty [53]. Our results suggest that transitioning from an idealised Zr-4 wall to a realistic steel wall results in a TBR penalty of approximately 0.1 to 0.15.

6.2.2. Safety and Coolant Selection

The choice of coolant represents a stark trade-off between performance and safety. Water (H_2O) was shown to enhance TBR (Figure 7) significantly better than gaseous coolants due to its moderating power, which softens the neutron spectrum into the thermal region where the ^{6}Li cross-section is largest (Figure 2). However, the use of water in a lithium-based blanket introduces the risk of violent exothermic chemical reactions in the event of a leak [54, 51]. While helium avoids this chemical reactivity, our results show it acts effectively as a void, increasing neutron leakage and degrading TBR [21]. A realistic design must therefore weigh the significant neutronic advantage of water moderation against the engineering complexity of double-walled piping and pressure suppression systems required to mitigate chemical hazards [19].

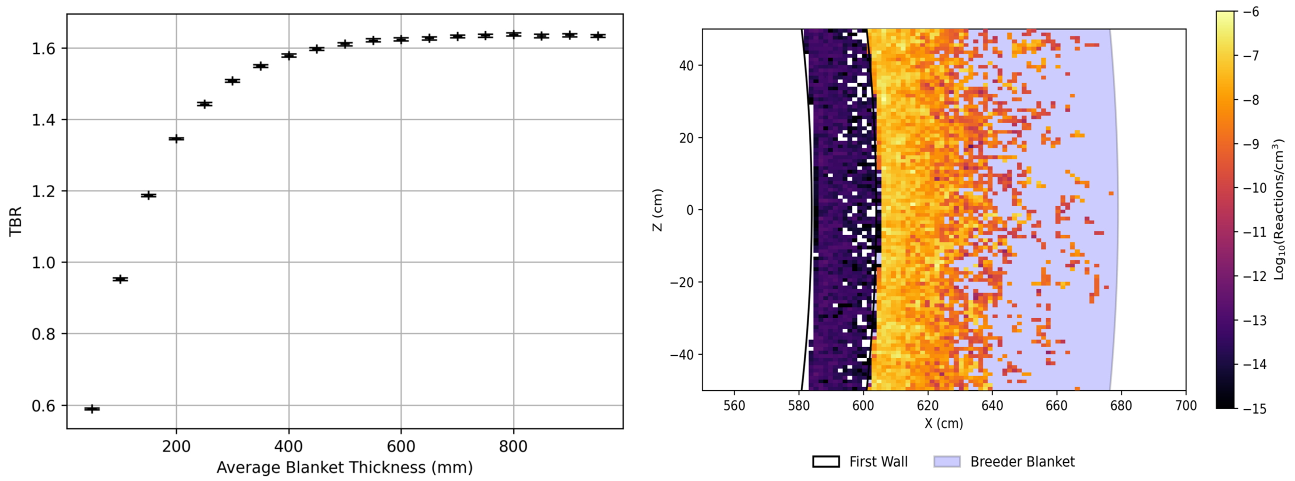

Figure 10: Analysis of breeder blanket radial thickness. On the left, TBR is plotted against average blanket thickness. TBR saturates at approximately 500 mm, suggesting full neutron attenuation. The cross-sectional heatmap of reaction density is plotted on the right. The reaction rate drops to negligible levels beyond the $640 - 580 = 60$cm depth into the blanket, confirming the saturation thickness observed from the left graph. The vast majority of the reactions occur near the surface, up to $\sim$ 20cm depth.

6.2.3. Energy Extraction (Heating)

A counter-intuitive trade-off exists regarding nuclear heating. While $Li_{17}Pb_{83}$ provided the highest TBR, it exhibited lower average heating per neutron compared to ceramic breeders (Table 1 vs Table 2). This is intrinsic to the physics of the multiplier: the $(n, 2\,n)$ reaction in lead is endothermic, consuming incident energy to liberate neutrons [29]. However $Li_{17}Pb_{83}$ has another draw: being eutectic also reduces the melting point (100°C lower), which draws significantly less energy in operation [48]. In contrast, the parasitic absorption in molten salts yields high heating but low breeding [33]. Solid breeders provide the best heating, and decent TBR. For a commercial power plant, where electricity generation is as vital as fuel self-sufficiency, the lower energy multiplication factor of lead-lithium systems is a distinct disadvantage that must be factored into the overall plant efficiency calculations [20].

6.3. Model Limitations

The primary limitation of this study lies in the homogenisation of the blanket materials. By treating the breeder and coolant as a uniform mixture, the simulation neglects "streaming effects", where neutrons escape through the void spaces of coolant channels without interaction [27]. This likely results in a slight overestimation of the TBR in our results compared to a heterogeneous model. The TBR values here do often exceed literature [20, 49, 42, 43]

Furthermore, the negligible impact of triangularity on TBR (Figure 8) is likely an artifact of the simplified neutron source definition. The model utilised a uniform ring source which does not geometrically conform to the plasma shape variations associated with high triangularity [45]. In a physical tokamak, increasing triangularity compresses the plasma volume and alters the view factor of the neutrons incident on the first wall, an effect not captured by the current source definition [55].

6.4. Next Steps

Future work will focus on increasing the fidelity of the simulation model to address the limitations identified above.

- **Heterogeneous Blanket Geometry:** Utilising the advanced features of Paramak to model discrete cooling channels and pebble beds to quantify the effects of neutron streaming [45]. Additionally, accounting for packing factors for Li_xPb_{100-x} alloys could provide a more informed sweep of neutron multiplier ratios.
- **Plasma Source Definition:** Replacing the ring source with a parametric plasma source that accurately maps the spatial distribution of fusion events, allowing for a true investigation into the effects of plasma geometry [56].
- **Structural Integration:** Moving beyond the simplified structure of the few most relevant neutronics layers, and instead including a more complex structure including key components like the external toroidal field coils. This will provide a more conservative and realistic estimate of the global TBR [18, 37].

Beyond increasing model fidelity, future studies should adopt a more holistic design approach that optimises other critical criteria alongside TBR, such as nuclear heating or safety considerations. Coupled parameter sweep studies offer a robust framework for this multi-objective optimisation. A particular focus should be placed on minimising leakage fractions; ensuring leakage approaches zero is essential for maximising the neutron economy available for breeding and heating, while simultaneously mitigating safety risks associated with external neutron radiation. Tracking this parameter is therefore recommended to ensure global design optimality.

7. CONCLUSION

This project utilised a parametric Paramak-OpenMC workflow to optimise tokamak breeder blankets for tritium self-sufficiency. By quantifying the impact of material composition and geometry, this study identified the critical design constraints required to realise a TBR greater than unity for practical devices, when accounting for additional breeding inefficiency in practice.

The eutectic alloy $Li_{17}Pb_{83}$ emerged as the optimal breeder, achieving a maximum TBR of 1.61. This superior performance results from the high packing density of the eutectic phase and effective $(\mathrm{n}, 2\,\mathrm{n})$ multiplication in lead, significantly outperforming molten salt concepts while also boasting lower melting points. Solid breeders are still a strong practical alternative due to better heat retention, no MHD effects, easier tritium extraction and a strong TBR of 1.48 ± 0.03 under our base reactor setup.

This study overturns the assumption that maximal enrichment is required; results indicate TBR saturation at 40% ^{6}Li. Enrichment beyond this threshold degrades performance by eliminating the neutron-preserving $^7\mathrm{Li}(\mathrm{n}, \mathrm{n}'\alpha)\mathrm{t}$ reaction, suggesting that commercial reactors can operate efficiently with moderate enrichment.

Geometric parameter sweeps determined a saturation blanket thickness of 500–600 mm, governed by the neutron mean free path. Notably, water coolant was found to enhance breeding via spectral moderation, whereas gaseous coolants increased leakage, presenting a clear trade-off between neutronic efficiency and chemical safety.

Ultimately, while these results establish robust trends, the reliance on homogenised materials and simplified sources likely overestimates TBR. Future work must therefore transition to high-fidelity heterogeneous models—incorporating discrete cooling channels and realistic plasma sources—to provide the conservative predictions necessary for the engineering design of next-generation fusion devices.

REFERENCES

Note: DOI links are hyperlinked within "[DOI]" placeholder

1. Lilley J. Nuclear Physics: Principles and Applications. Manchester Physics Series. John Wiley and Sons, 2008
2. Heckrotte W, Hiskes J. Some factors in the choice of D-D, D-T or D-3He mirror fusion power systems. Nuclear Fusion. 1971 Oct; 11:471–84. [DOI]
3. Brown D, Chadwick M, Capote R, Kahler A, Trkov A, Herman M, et al. ENDF/B-VIII.0: The 8 th Major Release of the Nuclear Reaction Data Library with CIELO-project Cross Sections, New Standards and Thermal Scattering Data. Nuclear Data Sheets. 2018 Feb; 148:1–142. [DOI]
4. Sawan M, Abdou M. Physics and technology conditions for attaining tritium self-sufficiency for the DT fuel cycle. Fusion Engineering and Design. 2006 Feb; 81:1131–44. [DOI]
5. Federici G, Biel W, Gilbert M, Kemp R, Taylor N, Wenninger R. European DEMO design strategy and consequences for materials. Nuclear Fusion. 2017 Sep 1; 57:1–26. [DOI]
6. Lord M, Bennett I, Harrington C, Cooper A, Lee-Lane D, Cureton A, et al. Fusing together an outline design for sustained fuelling and tritium self-sufficiency. Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences. 2024 Oct 9; 382:1–15. [DOI]
7. Wesson J. Tokamaks. 4th. Oxford University Press, 2011
8. Shatalov G, Abdou M, Antipenkov A, Daenner W, Gohar Y, Kuroda T, et al. Breeder and test blankets in ITER. Fusion Engineering and Design. 1991 Dec; 16:85–93. [DOI]
9. Hernández F, Pereslavtsev P. First principles review of options for tritium breeder and neutron multiplier materials for breeding blankets in fusion reactors. Fusion Engineering and Design. 2018 Dec; 137:243–56. [DOI]
10. Kuridan RM. Neutron Transport: Theory, Modeling, and Computations. Graduate Texts in Physics. Cham: Springer Nature Switzerland, 2023. [DOI]
11. Tas H, Malang S, Reiter F, Sannier J. Liquid breeder materials. Journal of Nuclear Materials. 1988 Jul; 155-157:178–87. [DOI]
12. Paris MW, Chadwick MB. A lost detail in D–T fusion history. Physics Today. 2023 Oct 1; 76:10–11. [DOI]
13. Kovari M, Coleman M, Cristescu I, Smith R. Tritium resources available for fusion reactors. Nuclear Fusion. 2018 Feb 1; 58:1–10. [DOI]
14. Pearson RJ, Antoniazzi AB, Nuttall WJ. Tritium supply and use: a key issue for the development of nuclear fusion energy. Fusion Engineering and Design. 2018 Nov; 136:1140–8. [DOI]
15. Moir RW. The fusion breeder. Journal of Fusion Energy. 1982 Oct; 2:351–67. [DOI]
16. Lee JD. U.S.-DOE Fusion-Breeder Program: Blanket Design and System Performance. Proceedings of the Third International Conference on Emerging Nuclear Energy Systems. 1983 :1–37
17. Maniscalco JA, Berwald DH, Moir RW, Lee JD, Teller E. The Fusion Breeder — an Early Application of Nuclear Fusion. Fusion Technology. 1984 Nov; 6:584–96. [DOI]
18. Federici G, Boccaccini L, Cismondi F, Gasparotto M, Poitevin Y, Ricapito I. An overview of the EU breeding blanket design strategy as an integral part of the DEMO design effort. Fusion Engineering and Design. 2019 Apr; 141:30–42. [DOI]
19. Liu S, Chen L, Ma X, Cheng X, Lu P, Lei M, et al. Design of the Water-Cooled Ceramic Breeder blanket for CFETR. Fusion Engineering and Design. 2022 Apr; 177:1–8. [DOI]
20. Meschini S, Ferry SE, Delaporte-Mathurin R, Whyte DG. Modeling and analysis of the tritium fuel cycle for ARC- and STEP-class D-T fusion power plants. Nuclear Fusion. 2023 Dec 1; 63:1–34. [DOI]

21. Wang X, Feng K, Chen Y, Zhang L, Feng Y, Wu X, et al. Current design and R&D progress of the Chinese helium cooled ceramic breeder test blanket system. Nuclear Fusion. 2019 Jul 1; 59:1–8. [DOI]

22. Sorbom B, Ball J, Palmer T, Mangiarotti F, Sierchio J, Bonoli P, et al. ARC: A compact, high-field, fusion nuclear science facility and demonstration power plant with demountable magnets. Fusion Engineering and Design. 2015 Nov; 100:378–405. [DOI]

23. Shimwell J, Lilley S, Morgan L, Packer L, Kovari M, Zheng S, et al. Reducing beryllium content in mixed bed solid-type breeder blankets. Fusion Engineering and Design. 2016 Nov; 109-111:1564–8. [DOI]

24. Wu Y. Overview of liquid lithium lead breeder blanket program in China. Fusion Engineering and Design. 2011 Oct; 86:2343–6. [DOI]

25. Bocci B, Hartwig Z, Segantin S, Testoni R, Whyte D, Zucchetti M. ARC reactor materials: Activation analysis and optimization. Fusion Engineering and Design. 2020 May 1; 154:1–8. [DOI]

26. Chen H, Li M, Lv Z, Zhou G, Liu Q, Wang S, et al. Conceptual design and analysis of the helium cooled solid breeder blanket for CFETR. Fusion Engineering and Design. Proceedings of the 28th Symposium On Fusion Technology (SOFT-28) 2015 Oct 1; 96-97:89–94. [DOI]

27. Fradera J, Sádaba S, Calvo F, Ha S, Merriman S, Gordillo P, et al. Pre-conceptual design of an encapsulated breeder commercial blanket for the STEP fusion reactor. Fusion Engineering and Design. 2021 Nov; 172:1–15. [DOI]

28. Fernández-Berceruelo I, Palermo I, Urgorri F, Rapisarda D, González M, Alguacil J, et al. Progress in design and experimental activities for the development of an advanced breeding blanket. Nuclear Fusion. 2024 May 1; 64:1–16. [DOI]

29. Graham G, Pearce A, Morris J, Kembleton R. A Dual Coolant Lithium Lead Breeder Blanket for a Fusion Power Plant Systems Model. Journal of Nuclear Engineering. 2022; 1:1–8

30. Malang S, Deckers H, Fischer U, John H, Meyder R, Norajitra P, et al. Self-cooled blanket concepts using Pb-17Li as liquid breeder and coolant. Fusion Engineering and Design. 1991 Apr 2; 14:373–99. [DOI]

31. Morley NB, Malang S, Kirillov I. Thermofluid Magnetohydrodynamic Issues for Liquid Breeders. Fusion Science and Technology. 2005 Apr; 47:488–501. [DOI]

32. Segantin S, Testoni R, Hartwig Z, Whyte D, Zucchetti M. Optimization of tritium breeding ratio in ARC reactor. Fusion Engineering and Design. 2020 May; 154:1–5. [DOI]

33. Bohm TD, Lindley BA. Initial Neutronics Investigation of a Chlorine Salt–Based Breeder Blanket. Fusion Science and Technology. 2023 Nov 17; 79:995–1007. [DOI]

34. Proust E, Anzidei L, Dalle Donne M, Fischer U, Kuroda T. Solid breeder blanket design and tritium breeding. Fusion Engineering and Design. 1991 Dec; 16:73–84. [DOI]

35. Knitter R, Chaudhuri P, Feng Y, Hoshino T, Yu IK. Recent developments of solid breeder fabrication. Journal of Nuclear Materials. 2013 Nov; 442:S420–S424. [DOI]

36. Chaudhuri P, Danani C, Chaudhari V, Rajendra Kumar E. Overview of design and thermal–hydraulic analysis of Indian solid breeder blanket concept. Fusion Engineering and Design. 2013 May; 88:209–15. [DOI]

37. Lei M, Wu Q, Xu S, Xu K, Ma X, Liu S. Design and analysis of the equatorial inboard WCCB blanket module for CFETR. Fusion Engineering and Design. 2020 Dec; 161:1–9. [DOI]

38. Shi Y, Lu T, Gao T, Xiang X, Zhang Q, Yu X, et al. Density functional study of lithium vacancy in Li4SiO4: Trapping of tritium and helium. Journal of Nuclear Materials. 2015 Dec; 467:519–26. [DOI]

39. Delaporte-Mathurin R, Goles N, Ball J, Dunn C, Edwards E, Ferry S, et al. Advancing tritium self-sufficiency in fusion power plants: insights from the BABY experiment. Nuclear Fusion. 2025 Feb 1; 65:1–11. [DOI]

40. Romano PK, Horelik NE, Herman BR, Nelson AG, Forget B, Smith K. OpenMC: A state-of-the-art Monte Carlo code for research and development. Annals of Nuclear Energy. 2015 Aug; 82:90–7. [DOI]

41. Bae JW, Peterson E, Shimwell J. ARC reactor neutronics multi-code validation*. Nuclear Fusion. 2022 Jun 1; 62:1–17. [DOI]

42. Segantin S, Testoni R, Zucchetti M. Neutronic comparison of liquid breeders for ARC-like reactor blankets. Fusion Engineering and Design. 2020 Nov; 160:1–10. [DOI]

43. Goel V, Aslam S, Dua S. Optimization of Tritium Breeding Ratio in a DT and DD Submersion Tokamak Fusion Reactor. 2023 Sep 30. [DOI]

44. Kalos MH, Whitlock PA. Monte Carlo Methods. 1st ed. Wiley, 2008 Sep 17. [DOI]

45. Shimwell J, Billingsley J, Delaporte-Mathurin R, Morbey D, Bluteau M, Shriwise P, et al. The Paramak: automated parametric geometry construction for fusion reactor designs. F1000Research. 2021 Jan 18; 10:27. [DOI]

46. Muir DW. Sensitivity of neutron multigroup cross sections to thermal broadening of the fusion peak. CONF-740402-33 LA-UR-74-613. USA: Los Alamos Scientific Lab, 1974

47. Wandelt K, ed. Encyclopedia of interfacial chemistry: surface science and electrochemistry. Amsterdam, Netherlands Oxford, UK Cambidge, USA: Elsevier, 2018. 1 p.

48. Mas de les Valls E, Sedano LA, Batet L, Ricapito I, Aiello A, Gastaldi O, et al. Lead–lithium eutectic material database for nuclear fusion technology. Journal of Nuclear Materials. Heavy Liquid Metal Cooled Reactors and Related Technologies 2008 Jun 15; 376:353–7. [DOI]

49. Boullon R, Jaboulay JC, Aubert J. Molten salt breeding blanket: Investigations and proposals of pre-conceptual design options for testing in DEMO. Fusion Engineering and Design. 2021 Oct 1; 171:1–9. [DOI]

50. Kurasawa T, Takatsu H, Sato S, Mori S, Hashimoto T, Nakahira M, et al. Ceramic breeding blanket development for experimental fusion reactor in JAERI. Fusion Engineering and Design. 1995 Mar; 27:449–56. [DOI]

51. Hirose T, Guan W, Katagiri T, Wakasa A, Someya Y, Nakajima M, et al. Functional tests for water cooled ceramic breeder blanket system using full-scale mockups. Fusion Engineering and Design. 2024 Mar; 200:1–6. [DOI]

52. King D, Knowles A, Bowden D, Wenman M, Capp S, Gorley M, et al. High temperature zirconium alloys for fusion energy. Journal of Nuclear Materials. 2022 Feb; 559:1–27. [DOI]

53. Stornelli G, Di Schino A, Montanari R, Sgambetterra M, Testani C, Varone A. Ultra-fine grained EUROFER97 steel for nuclear fusion applications. Journal of Materials Research and Technology. 2024 Nov; 33:5075–87. [DOI]

54. Kawamura Y, Hirose T, Guan W, Miyoshi Y, Katagiri T, Wakasa A, et al. Overview of progress on water cooled ceramic breeder blanket in Japan. Fusion Engineering and Design. 2024 Apr; 201:1–7. [DOI]

55. Creely AJ, Greenwald MJ, Ballinger SB, Brunner D, Canik J, Doody J, et al. Overview of the SPARC tokamak. Journal of Plasma Physics. 2020 Oct; 86:1–25. [DOI]

56. Fable E, Angioni C, Siccinio M, Zohm H. Plasma physics for fusion reactor system codes: Framework and model code. Fusion Engineering and Design. 2018 May; 130:131–6. [DOI]

A. SUPPLEMENTARY NEUTRONICS THEORY

A.1. Elastic Scattering Kinematics

In an elastic collision with a nucleus of mass number A, the average logarithmic energy decrement per collision (ξ) is determined by the target mass [1]:

$$\xi = 1 + \frac{(A-1)^2}{2A} \ln\left(\frac{A-1}{A+1}\right) \tag{10}$$

For Hydrogen ($A \approx 1$), $\xi = 1$, indicating that a neutron can lose all its energy in a single head-on collision. For lead ($A \approx 208$), $\xi \approx 0.0096$, necessitating hundreds of collisions to achieve the same thermalisation.